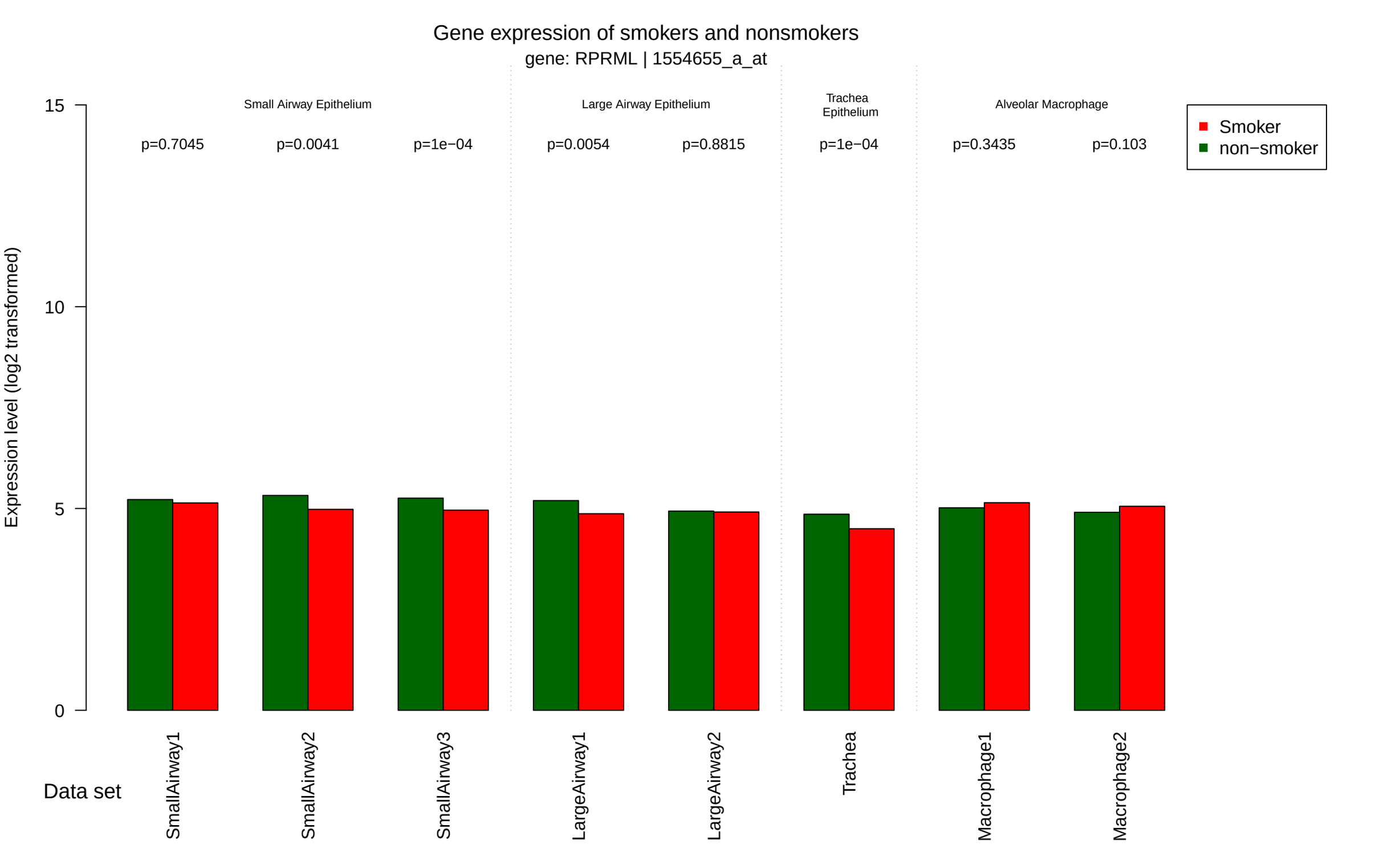
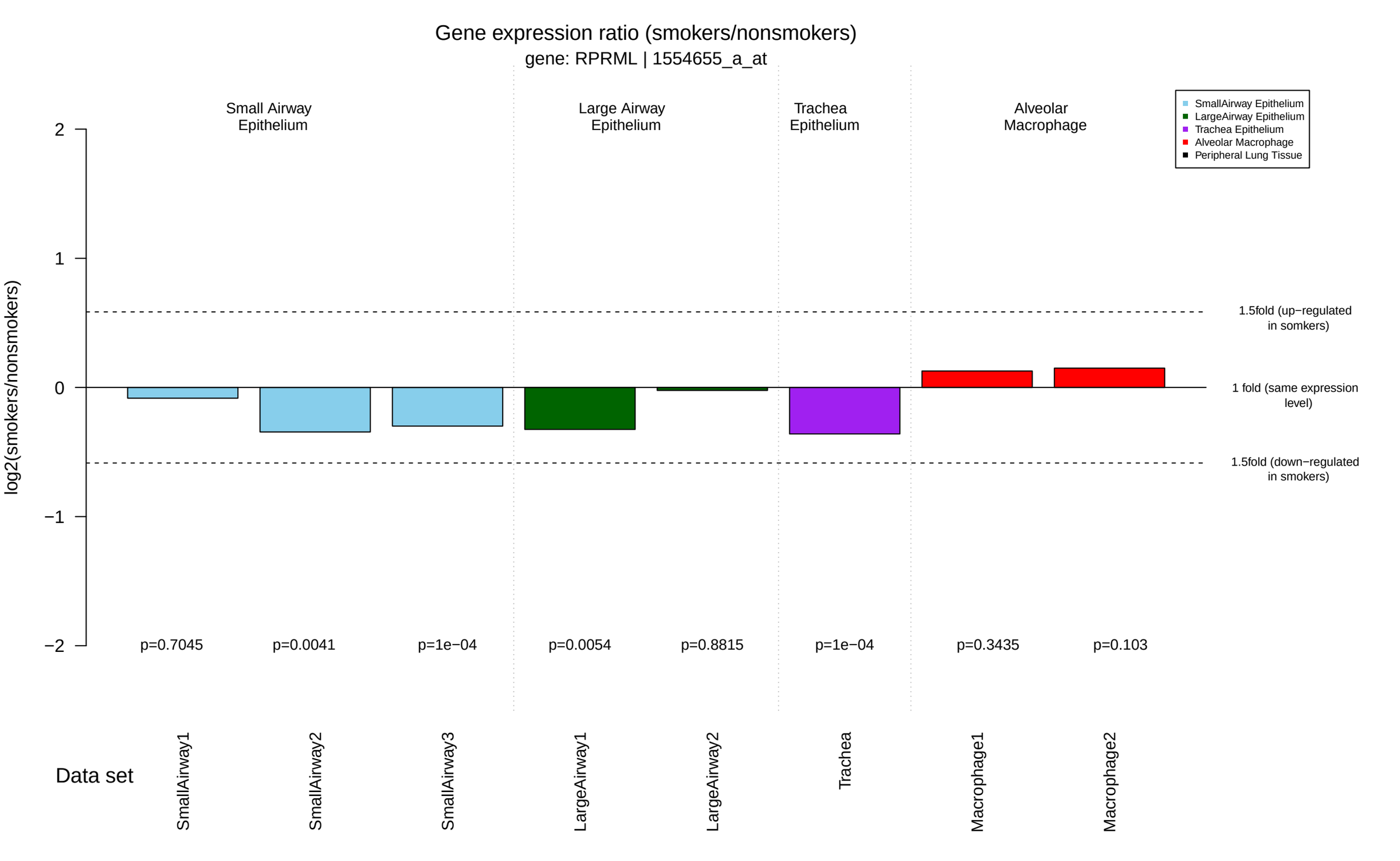
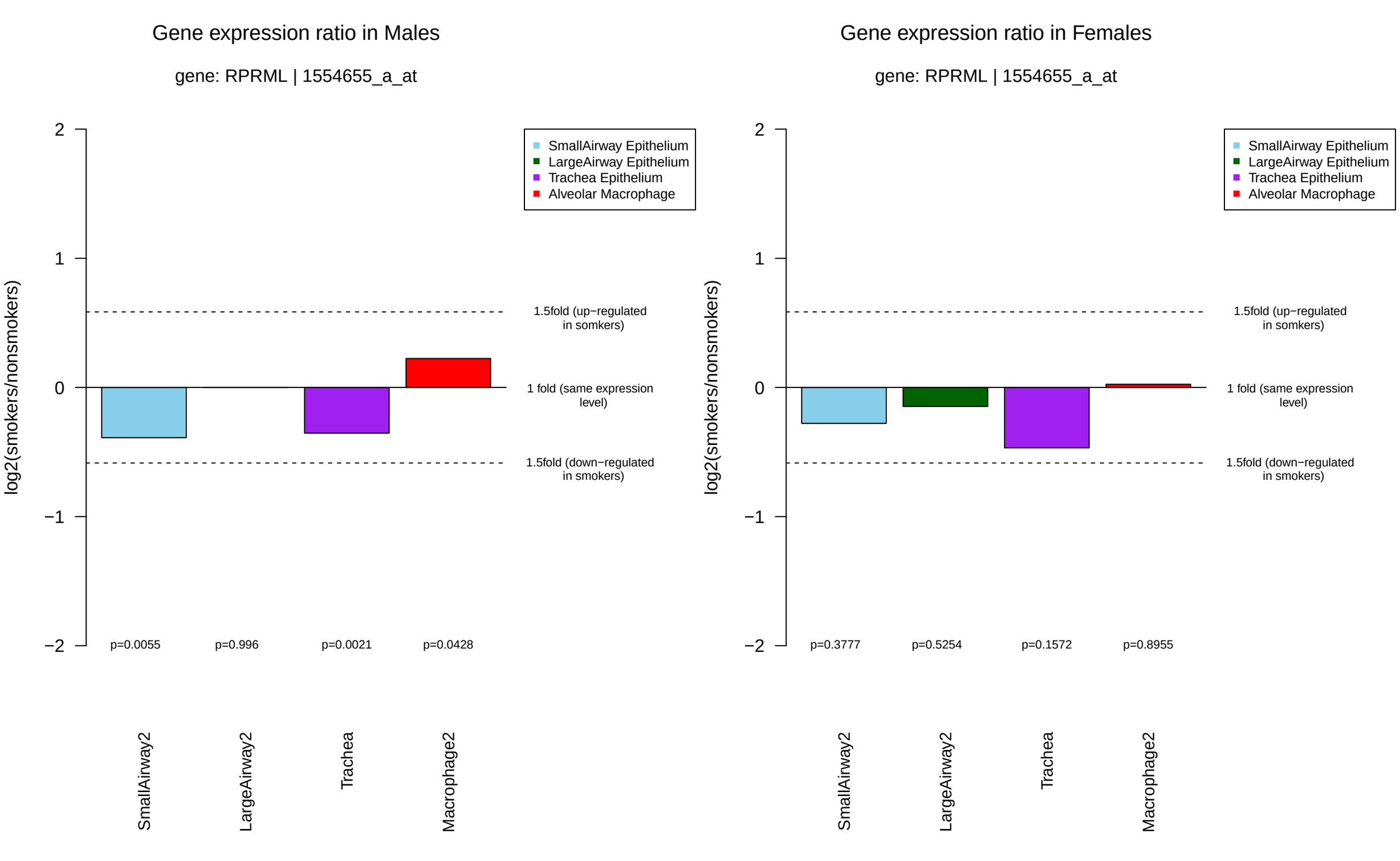
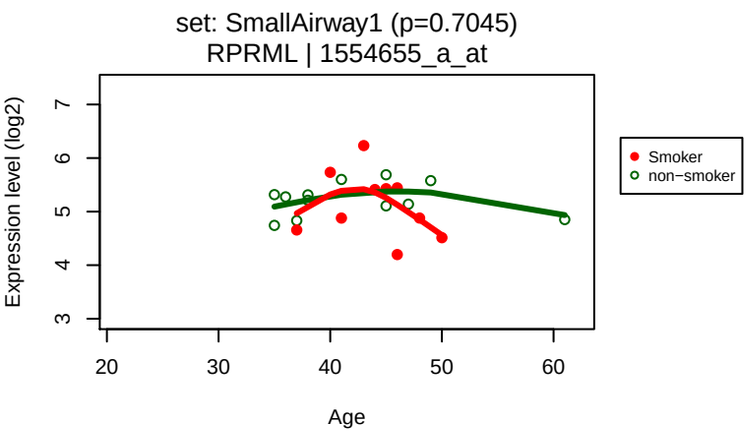
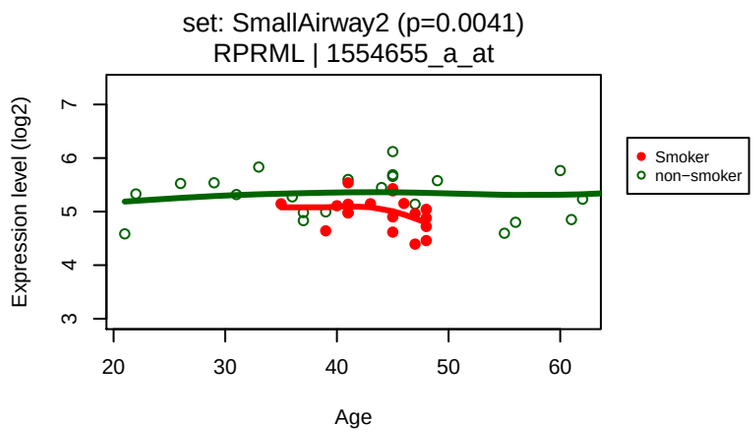
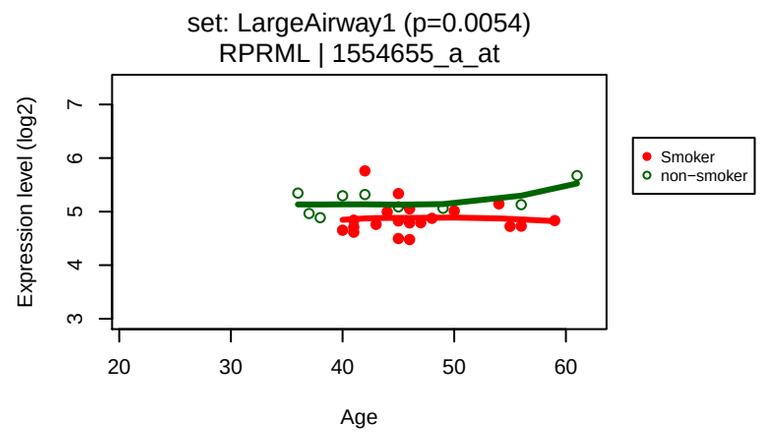

Gene expression of smokers and nonsmokers
gene: RPRML | 1554655_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7045
p=0.0041
p=1e−04
p=0.0054
p=0.8815
p=1e−04
p=0.3435
p=0.103
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

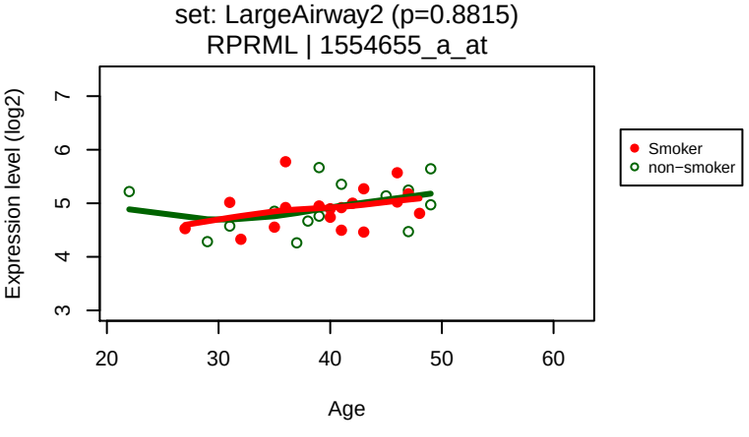

Gene expression ratio (smokers/nonsmokers)
gene: RPRML | 1554655_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7045
p=0.0041
p=1e−04
p=0.0054
p=0.8815
p=1e−04
p=0.3435
p=0.103
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

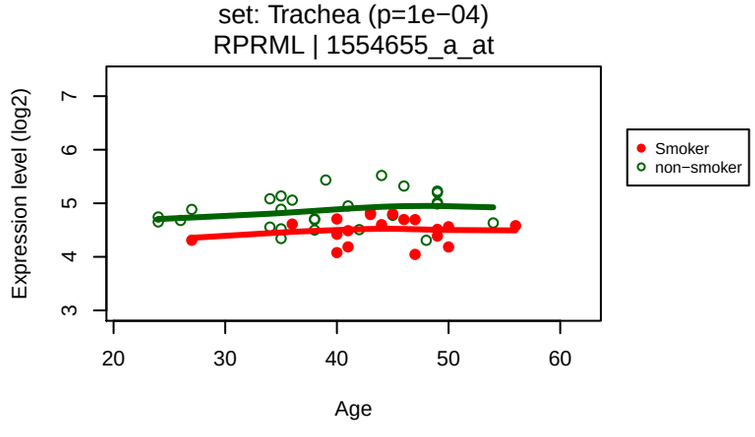

Gene expression ratio in Males
gene: RPRML | 1554655_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0055
p=0.996
p=0.0021
p=0.0428
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRML | 1554655_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3777
p=0.5254
p=0.1572
p=0.8955
SmallAirway2
LargeAirway2
Trachea
Macrophage2

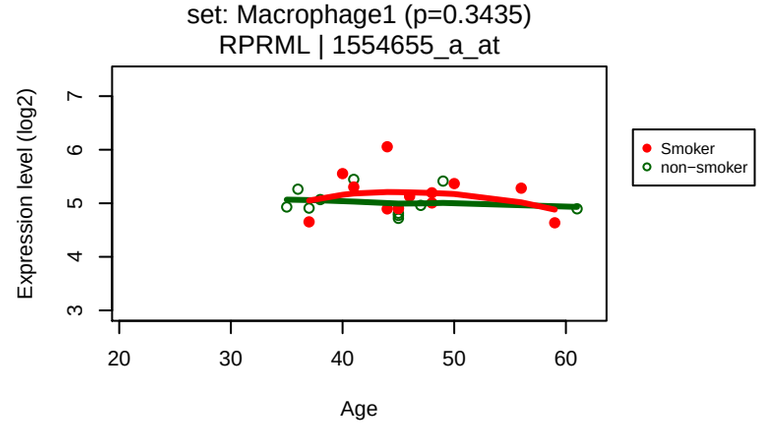

set: SmallAirway1 (p=0.7045)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker

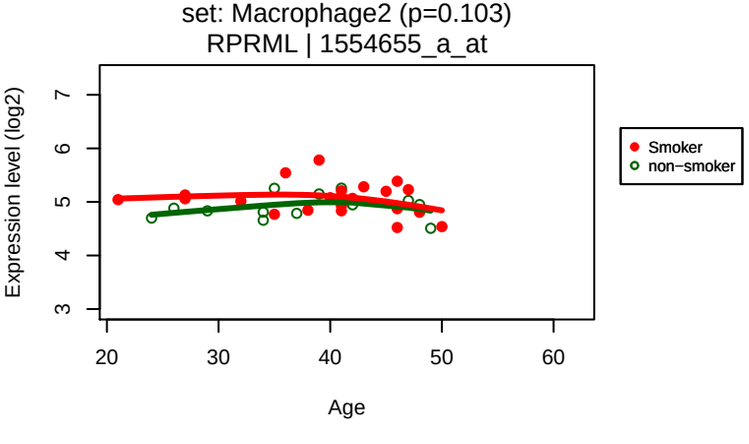

set: SmallAirway2 (p=0.0041)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker

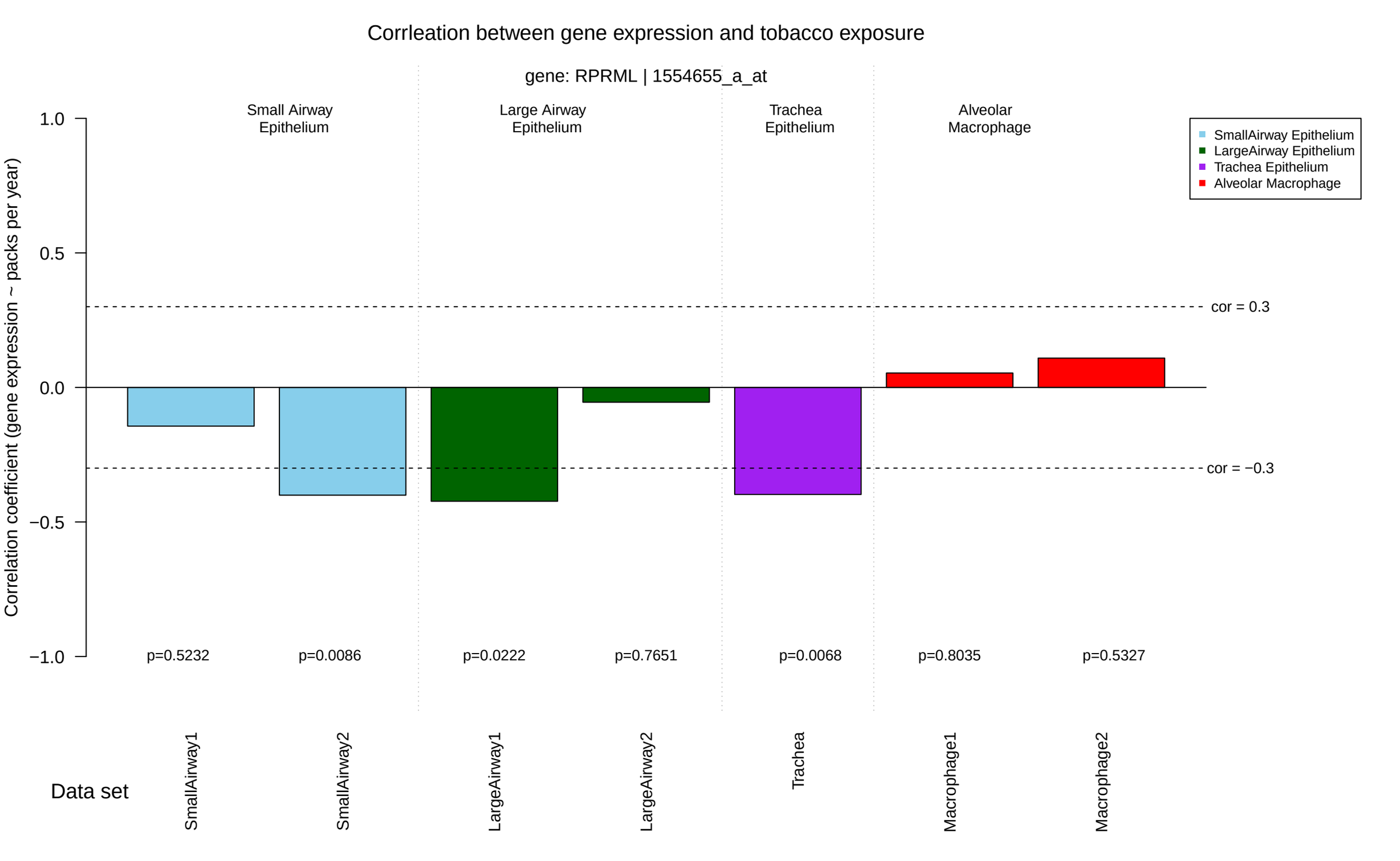

set: LargeAirway1 (p=0.0054)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8815)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3435)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.103)
RPRML | 1554655_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRML | 1554655_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5232
p=0.0086
p=0.0222
p=0.7651
p=0.0068
p=0.8035
p=0.5327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2